LAPORAN WORKSHOP DEVELOPER OPERASIONAL

LARAVEL DEPLOYMENT ON KUBERNETES (MINIKUBE)

MINGGU KE - 12

oleh

Beniqno Andi Priambudi Kusuma

NIM E41230557

GOLONGAN : B

PROGRAM STUDI TEKNIK INFORMATIKA / TEKNIK INFORMATIKA JEMBER

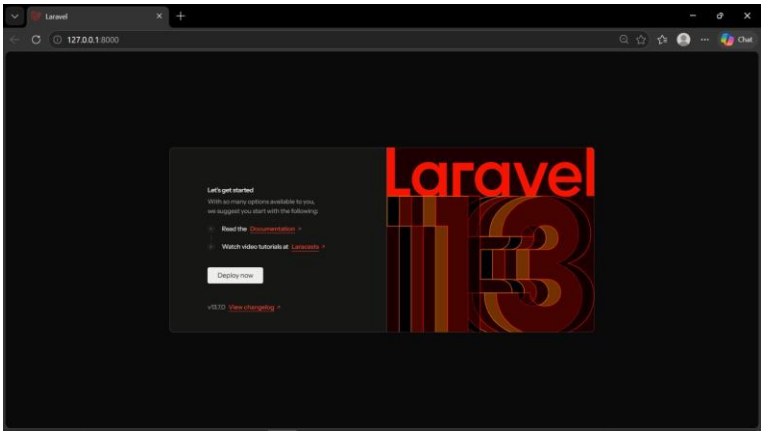
JURUSAN TEKNOLOGI INFORMASI

POLITEKNIK NEGERI JEMBER

2026

1. **Setup Proyek Laravel**

Pertama, membuat proyek Laravel yang akan di deploy ke Kubernetes nantinya. Untuk mengecek proyek Laravel berhasil dibuat dapat menjalankan Laravel secara local seperti di bawah ini.



2. **Konfigurasi Dockerfile**

Docker file berisi kode untuk instalasi image PHP 8.3 dengan Apache, lalu menginstal ekstensi MySQL, mengaktifkan modul rewrite URL, dan mengarahkan root dokumen ke folder public

```
.env    dockerfile
dockerfile
FROM php:8.3-apache

RUN docker-php-ext-install pdo pdo_mysql
RUN a2enmod rewrite

ENV APACHE_DOCUMENT_ROOT=/var/www/html/public

RUN sed -ri -e 's!/var/www/html!${APACHE_DOCUMENT_ROOT}!g' /etc/apache2/sites-available/000-default.c
RUN sed -ri -e 's!/var/www/html!${APACHE_DOCUMENT_ROOT}!g' /etc/apache2/apache2.conf

COPY --chown=www-data:www-data . /var/www/html

RUN chmod -R 775 /var/www/html/storage /var/www/html/bootstrap/cache

EXPOSE 80
```

3. **Inisialisasi Kubernetes (Minikube)**

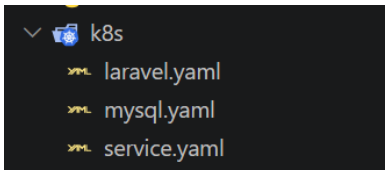
Perintah kubectl get nodes digunakan untuk memastikan cluster sudah siap. Jika muncul status "Ready", berarti minikube kita sudah bisa digunakan.

```
PS D:\Perkuliahan\Mata Kuliah\Project Aplikasi\laravel-minikube> kubectl get nodes
NAME       STATUS   ROLES           AGE   VERSION
minikube   Ready    control-plane   48m   v1.35.1
```

4. **Setup Kubernetes Manifest**

Manifest adalah file YAML yang berisi kode untuk Kubernetes. Kita buat tiga file,

a. **laravel.yaml** (berisi Deployment dan Service untuk Laravel),
b. **mysql.yaml** (berisi Deployment dan Service untuk database), dan
c. **service.yaml** (mendefinisikan cara mengakses aplikasi dari luar).



Setelah semua file siap, perintah *kubectl apply -f .* akan menjalankan kode manifest tersebut ke dalam cluster.

```
PS D:\Perkuliahan\Mata Kuliah\Project Aplikasi\laravel-minikube\k8s> kubectl get pods
NAME                        READY   STATUS    RESTARTS   AGE
laravel-575b8c55bf-6ffb2    1/1     Running   0          42m
laravel-575b8c55bf-ckh4b    1/1     Running   0          42m
laravel-575b8c55bf-kpnmv    1/1     Running   0          42m
mysql-5c4d589d75-cldm6      1/1     Running   0          42m
PS D:\Perkuliahan\Mata Kuliah\Project Aplikasi\laravel-minikube\k8s> kubectl get svc
NAME               TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
kubernetes         ClusterIP   10.96.0.1        <none>        443/TCP        45m
laravel            ClusterIP   10.96.225.24     <none>        80/TCP         42m
laravel-nodeport   NodePort    10.110.136.133   <none>        80:30007/TCP   42m
mysql              ClusterIP   10.107.72.193    <none>        3306/TCP       42m
```

Kemudian jalankan *kubectl get pods* yang berfungsi untuk melihat daftar semua pod (container aplikasi) yang sedang berjalan di Kubernetes dan

kubectl get svc berfungsi untuk melihat daftar layanan yang menjadi pintu akses masuk ke aplikasi, termasuk jenis service dan port yang digunakan.

5. Build dan Load Image ke Minikube

Dari Dockerfile yang sudah kita buat, kita membuat image bernama laravel-app menggunakan perintah *docker build*. Image ini masih ada di Docker Desktop, belum masuk ke Minikube, kita perlu memasukkan image ke minikube dengan perintah *image load*

```
PS D:\Perkuliahan\Mata Kuliah\Project Aplikasi\laravel-minikube> docker build -t laravel-app .
[+] Building 264.7s (13/13) FINISHED                                                docker:desktop-linux
 => [internal] load build definition from dockerfile                                                  0.1s
 => => transferring dockerfile: 487B                                                                  0.1s
 => [internal] load metadata for docker.io/library/php:8.3-apache                                     2.4s
 => [auth] library/php:pull token for registry-1.docker.io                                            0.0s
 => [internal] load .dockerignore                                                                     0.1s
 => => transferring context: 2B                                                                       0.0s
 => [1/7] FROM docker.io/library/php:8.3-apache@sha256:b0e30bc07e65cedc7cb5563f5637d29d04d6fa3e41   164.5s
 => => resolve docker.io/library/php:8.3-apache@sha256:b0e30bc07e65cedc7cb5563f5637d29d04d6fa3e4109   0.0s
 => => sha256:0f7cba4685aaef664e0a9b25c22f49e5917e5968d70e472e6e40eb661c1721a8 890B                   0.7s
 => => sha256:f63167452cc5461edf55f1074b3d82f87c6cc942099c6f78e30166dbee600481 246B / 246B            1.1s
 => => sha256:fe11d9786149bd90ddcdf5bfcbd4a7b7728fe454befff9d40df5d9164a704f85 251B / 251B            1.2s
 => => sha256:418def1552525b1a6ff8e44a8b86ed54535dda3170ce80ea29357a8869e67a20 2.46kB / 2.46kB        1.4s
 => => sha256:6ac490a3731b7bd2e09c6e56a9d2e84259101aa6fd71c4af241788ee652493a0 11.72MB / 11.72MB     32.0s
 => => sha256:c881ca0ad0b30cbd1db4cf96d5f769e7791b83398bdb03504c477284b82a5609 487B / 487B            1.3s
 => => sha256:2c9891ff646dc3cd3af1829b226bff68ad163d74241eb65c4bcab9064d76816b 12.77MB / 12.77MB     32.9s
 => => sha256:c5b991a71c26d784ac1b1e618b419cd829d52bd5c4ed4ecd9e90c655cefa9ef3 481B / 481B            1.6s
 => => sha256:b176a0029abea5d062fb7bb431cc2a090801dfefee0f89897bdc2446bffc971e 429B / 429B            1.2s
 => => sha256:a3c817c88db8e01c9e9b3d8f20bfd8dc273d3086fd8057736bb7545317adaca4 4.23MB / 4.23MB       23.4s
 => => sha256:a30254b6b683f3ad7b929cb5357e57af7c88937022c26a3cfb6047c15717ca90 225B / 225B            0.8s
 => => sha256:456e68dcd0176c22b4f231707613b3ad35220379a73572d9ae57eee3e5e8d7c 117.84MB / 117.84MB   148.8s
 => => sha256:7f7e820a0db114690e85203bc3f459f4138e78ea454d47578ef651c7b2524c3f 226B / 226B            0.8s
 => => extracting sha256:7f7e820a0db114690e85203bc3f459f4138e78ea454d47578ef651c7b2524c3f             0.1s
 => => extracting sha256:456e68dcd0176c22b4f231707613b3ad35220379a73572d9ae57eee3e5e8d7cb            10.2s
 => => extracting sha256:a30254b6b683f3ad7b929cb5357e57af7c88937022c26a3cfb6047c15717ca90             0.1s
 => => extracting sha256:a3c817c88db8e01c9e9b3d8f20bfd8dc273d3086fd8057736bb7545317adaca4             0.7s
 => => extracting sha256:b176a0029abea5d062fb7bb431cc2a090801dfefee0f89897bdc2446bffc971e             0.2s
 => => extracting sha256:c5b991a71c26d784ac1b1e618b419cd829d52bd5c4ed4ecd9e90c655cefa9ef3             0.0s
 => => extracting sha256:2c9891ff646dc3cd3af1829b226bff68ad163d74241eb65c4bcab9064d76816b             0.2s
 => => extracting sha256:c881ca0ad0b30cbd1db4cf96d5f769e7791b83398bdb03504c477284b82a5609             0.1s
 => => extracting sha256:6ac490a3731b7bd2e09c6e56a9d2e84259101aa6fd71c4af241788ee652493a0             0.0s
 => => extracting sha256:418def1552525b1a6ff8e44a8b86ed54535dda3170ce80ea29357a8869e67a20             0.1s
 => => extracting sha256:fe11d9786149bd90ddcdf5bfcbd4a7b7728fe454befff9d40df5d9164a704f85             0.0s
 => => extracting sha256:f63167452cc5461edf55f1074b3d82f87c6cc942099c6f78e30166dbee600481             0.0s
 => => extracting sha256:0f7cba4685aaef664e0a9b25c22f49e5917e5968d70e472e6e40eb661c1721a8             0.0s
```

```
PS D:\Perkuliahan\Mata Kuliah\Project Aplikasi\laravel-minikube> minikube image load laravel-app
```

Kemudian untuk masuk ke node minikube kita jalankan *minikube ssh* seperti gambar di bawah ini.

```
PS D:\Perkuliahan\Mata Kuliah\Project Aplikasi\laravel-minikube> minikube ssh
Linux minikube 6.6.87.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025 x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
docker@minikube:~$ exit
logout
```

Selanjutnya, untuk mengecek apakah image Laravel app latest telah dibuat dapat menjalankan perintah *docker image*.

```
PS D:\Perkuliahan\Mata Kuliah\Project Aplikasi\laravel-minikube> docker images
                                                                                                  i Info →   U  In Use
IMAGE                                                                                   ID             DISK USAGE   CONTENT SIZE   EXTRA
alpine:latest                                                                           5b10f432ef3d       13.1MB         3.95MB
docker/desktop-kubernetes:kubernetes-v1.34.1-cni-v1.7.1-critools-v1.33.0-cri-dockerd-v0.3.20-1-debian
                                                                                        12d6673564e0        592MB          185MB
docker/desktop-storage-provisioner:v3.0                                                 57d2b6ad1c6f       80.3MB         23.9MB
docker/desktop-vpnkit-controller:v4.0                                                   bdaff3408b1c       52.6MB         11.9MB
gcr.io/distroless/static-debian11:latest                                                1dbe426d60ca       8.23MB         1.01MB
gcr.io/k8s-minikube/kicbase:v0.0.50                                                     b97074569ae9       1.94GB          545MB    U
gcr.io/k8s-minikube/kicbase@sha256:eb4fec00e8ad70adf8e6436f195cc429825ffb85f95afcdb5d8d9deb576f3e93
                                                                                        eb4fec00e8ad       1.94GB          545MB
java-app:latest                                                                         f6d36ba1b056        610MB          193MB
laravel-app:latest                                                                      34c059235299        825MB          201MB
laravel-docker-app:latest                                                               6d612325a707       1.31GB          285MB
mysql:5.7                                                                               4bc6bc963e6d        700MB          149MB    U
nginx:alpine                                                                            1d13701a5f9f       93.4MB         26.9MB    U
nginx:latest                                                                            6e23479198b9        240MB         65.8MB
node-app:latest                                                                         9d2e6308fdcf        180MB         42.9MB
php-app:latest                                                                          86c79d7b3495        707MB          176MB
phpmyadmin/phpmyadmin:latest                                                            42a200db07b4       1.09GB          221MB    U
registry.k8s.io/coredns/coredns:v1.12.1                                                 e8c262566636        101MB         22.4MB
registry.k8s.io/etcd:3.6.4-0                                                            e36c08168342        273MB         74.3MB
registry.k8s.io/ingress-nginx/controller@sha256:e24f39d3eed6bcc239a56f2009887884 5f62baa34b9f2be2fd2c38ce9fb0f29e
                                                                                        e24f39d3eed6        392MB          104MB
registry.k8s.io/ingress-nginx/kube-webhook-certgen@sha256:36d05b4077fb8e3d13663702fa337f124675ba8667cbd949c03a8e8ea6fa4366
                                                                                        36d05b4077fb       83.9MB         23.9MB
registry.k8s.io/kube-apiserver:v1.34.1                                                  b9d7c117f8ac        118MB         27.1MB
registry.k8s.io/kube-controller-manager:v1.34.1                                         2bf47c1b01f5        101MB         22.8MB
registry.k8s.io/kube-proxy:v1.34.1                                                      913cc83ca0b5        102MB           26MB
registry.k8s.io/kube-scheduler:v1.34.1                                                  6e9fbc4e25a5       73.5MB         17.4MB
registry.k8s.io/metrics-server/metrics-server:v0.8.1                                    b2d2efaf5ac3        109MB         22.5MB
registry.k8s.io/pause:3.10                                                              ee6521f290b2       1.06MB          318kB
registry.k8s.io/pause:3.10.1                                                            278fb9dbcca9       1.06MB          318kB
```

6. **Konfigurasi env**

Pada env Laravel bagian database ubah db host dari port local menjadi mysql agar dapat terhubung dengan MySQL yang ada pada Kubernetes.

```
DB_CONNECTION=mysql
DB_HOST=mysql
DB_PORT=3306
DB_DATABASE=laravel_k8s
DB_USERNAME=root
DB_PASSWORD=
```

7. **Menjalankan Aplikasi Laravel di Kubernetes**

Sebelum menjalankan Laravel, kita perlu migrasi database dulu ke Laravel di Kubernetes dengan cara masuk ke dalam pod Laravel dengan perintah kubectl exec -it (pod Laravel) – bash.

```
PS D:\Perkuliahan\Mata Kuliah\Project Aplikasi\laravel-minikube> kubectl get pods
NAME                      READY   STATUS    RESTARTS        AGE
laravel-575b8c55bf-6ffb2  1/1     Running   1 (4m17s ago)   46h
laravel-575b8c55bf-ckh4b  1/1     Running   1 (4m17s ago)   46h
laravel-575b8c55bf-kpnmv  1/1     Running   1 (4m17s ago)   46h
mysql-5c4d589d75-cldm6    1/1     Running   1 (4m17s ago)   46h
```

Setelah masuk kita lakukan php artisan migrate untuk migrasi database.

```
PS D:\Perkuliahan\Mata Kuliah\Project Aplikasi\laravel-minikube> kubectl exec -it laravel-575b8c55bf-6ffb2
  -- bash
root@laravel-575b8c55bf-6ffb2:/var/www/html# php artisan migrate

  INFO  Preparing database.


  INFO  Running migrations.

  0001_01_01_000000_create_users_table ..................................................... 242.21ms DONE
  0001_01_01_000001_create_cache_table ..................................................... 146.46ms DONE
  0001_01_01_000002_create_jobs_table ...................................................... 199.15ms DONE
```

Lalu, untuk menjalankan Laravel pada Kubernetes kita dapat menjalankan perintah minikube *service laravel-nodeport* seperti di bawah ini.

```
PS D:\Perkuliahan\Mata Kuliah\Project Aplikasi\laravel-minikube> minikube service laravel-nodeport
```

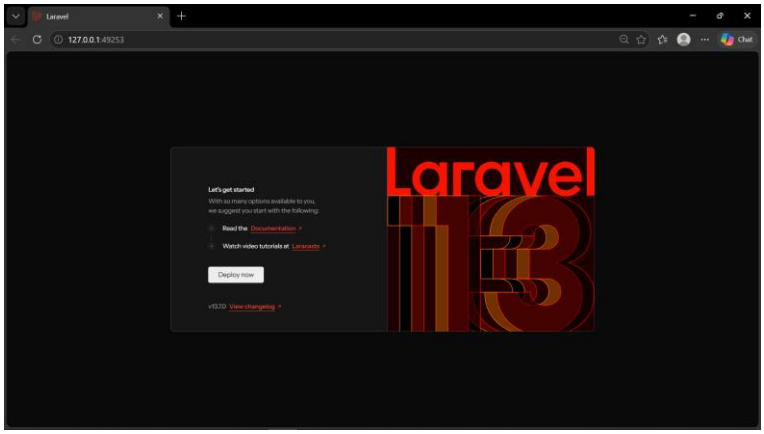
NAMESPACE	NAME	TARGET PORT	URL
default	laravel-nodeport	80	http://192.168.58.2:30007

```
🔗 Starting tunnel for service laravel-nodeport.
```

NAMESPACE	NAME	TARGET PORT	URL
default	laravel-nodeport		http://127.0.0.1:49253

```
🏃 Opening service default/laravel-nodeport in default browser...
❗ Because you are using a Docker driver on windows, the terminal needs to be open to run it.
```

Secara otomatis setelah menjalankan perintah tersebut akan muncul Laravel pada browser.



8. Manajemen Skalabilitas

Kita bisa menambah jumlah pod Laravel dari 2 menjadi 5 (scale up) atau sebaliknya (scale down). Kubernetes akan otomatis membuat pod baru atau menghapus pod lama

a. Replikasi 5 node

```
PS D:\Perkuliahan\Mata Kuliah\Project Aplikasi\laravel-minikube> kubectl scale deployment laravel --replic
as=5
deployment.apps/laravel scaled
PS D:\Perkuliahan\Mata Kuliah\Project Aplikasi\laravel-minikube> kubectl get pods -w
NAME                       READY   STATUS    RESTARTS        AGE
laravel-575b8c55bf-6ffb2   1/1     Running   1 (7m35s ago)   46h
laravel-575b8c55bf-8xgzn   1/1     Running   0               11s
laravel-575b8c55bf-ckh4b   1/1     Running   1 (7m35s ago)   46h
laravel-575b8c55bf-kpnmv   1/1     Running   1 (7m35s ago)   46h
laravel-575b8c55bf-n7664   1/1     Running   0               11s
mysql-5c4d589d75-cldm6     1/1     Running   1 (7m35s ago)   46h
PS D:\Perkuliahan\Mata Kuliah\Project Aplikasi\laravel-minikube> kubectl get deployment laravel
NAME      READY   UP-TO-DATE   AVAILABLE   AGE
laravel   5/5     5            5           46h
```

b. Replikasi 2 node

```
PS D:\Perkuliahan\Mata Kuliah\Project Aplikasi\laravel-minikube> kubectl scale deployment laravel --replic
as=2
deployment.apps/laravel scaled
PS D:\Perkuliahan\Mata Kuliah\Project Aplikasi\laravel-minikube> kubectl get pods -w
NAME                       READY   STATUS    RESTARTS      AGE
laravel-575b8c55bf-ckh4b   1/1     Running   1 (9m ago)    46h
laravel-575b8c55bf-kpnmv   1/1     Running   1 (9m ago)    46h
mysql-5c4d589d75-cldm6     1/1     Running   1 (9m ago)    46h
PS D:\Perkuliahan\Mata Kuliah\Project Aplikasi\laravel-minikube> kubectl get deployment laravel
NAME      READY   UP-TO-DATE   AVAILABLE   AGE
laravel   2/2     2            2           46h
```